ChemoInteract

Complete Documentation — Part 3 of 3

Interview Q&A, College Viva Preparation, Presentation Scripts, Resume Roadmap

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Framework:	PyTorch 2.0+	**Focus:**	Interview & Viva Prep

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1 Interview Preparation Guide

1.1 HR & Project Overview Questions

Q1: Tell me about your final year project.

Model Answer

"I developed ChemoInteract, a modular deep learning Python library for predicting drug-drug interactions. Adverse drug reactions cause thousands of preventable hospitalizations annually in polypharmacy patients. ChemoInteract unifies 11 published deep learning architectures — ranging from simple FC networks to Graph Neural Networks and cross-drug Attention models — under a single plug-and-play training pipeline. A researcher can benchmark any model across 5 auto-downloaded datasets with a single line of code."

Q2: What was your biggest technical challenge?

Model Answer

"Making 11 models with completely different input requirements work through one unified pipeline. Some take fingerprint vectors; others take molecular graphs. Some need cell line context; others do not. I solved this by implementing the **Template Method** pattern via an abstract `unpack()` method on the `Model` base class. Each model declares what it extracts from the `DrugPairBatch`, keeping the pipeline completely decoupled from model internals."

1.2 Technical Questions Model Answers

Q3: Why use ROC-AUC instead of Accuracy?

Model Answer

"Drug interaction datasets are heavily class-imbalanced (far more non-interacting pairs than interacting ones). A naive model predicting all zeros gets high accuracy but is clinically useless. ROC-AUC measures the probability that a randomly chosen positive pair receives a higher prediction score than a randomly chosen negative pair, making it threshold-independent and robust to class imbalance."

Q4: Explain `@lru_cache` in your data loader.

Model Answer

"`@lru_cache(maxsize=1)` memoises dataset loading methods. The first call downloads and parses data from GitHub/disk; subsequent calls return the cached in-memory objects instantly, avoiding redundant network I/O during repeated training runs."

2 College Viva Questions and Answers

Viva Question

V1: What does DDI stand for?

Drug-Drug Interaction: when the pharmacological effect of one drug is altered by another drug co-administered in the body.

Viva Question

V2: How many deep learning models are in ChemoInteract?

11 models: DeepSynergy, DeepDDI, MatchMaker (FC models); DeepDDS, DeepDrug, CASTER, GCNBMP, EPGCN-DS, MR-GNN (GNN models); MHCADDI, SSI-DDI (Attention models).

Viva Question

V3: What is a SMILES string?

Simplified Molecular Input Line Entry System: an ASCII text representation of chemical structures (e.g. Aspirin is `CC(=O)Oc1ccccc1C(=O)O`). TorchDrug parses SMILES strings into molecular graphs.

Viva Question

V4: What is a Morgan Fingerprint?

A 256-bit binary vector encoding the presence of circular chemical substructures up to radius 2 bonds around each atom.

Viva Question

V5: What does `model.eval()` do?

Disables Dropout (all neurons stay active) and locks Batch Normalization to use stored running statistics during evaluation.

Viva Question

V6: What is the Closed-World Assumption (CWA)?

Assuming that any unrecorded drug pair in database records is non-interacting (label = 0.0) during negative sampling.

3 Presentation Scripts

3.1 2-Minute Elevator Pitch

"I built ChemoInteract, a PyTorch library using deep learning to predict dangerous drug-drug interactions. Adverse drug reactions cause thousands of hospitalizations annually.

My primary contribution is unifying 11 published deep learning architectures — from basic neural networks to GNNs with cross-drug co-attention — into a single framework. Any model can be trained on any of 5 benchmark datasets with a single line of code.

The library automatically handles GPU acceleration, dataset downloading, and metric evaluation, achieving ROC-AUC scores up to 0.87 on benchmark datasets."

3.2 30-Second Pitch

"ChemoInteract is an open-source PyTorch library that uses Graph Neural Networks and AI to predict drug interactions and synergies. It unifies 11 published deep learning models across 5 datasets in a single line of code, enabling rapid computational drug discovery."

4 Resume Preparation

4.1 Resume Bullet Points

- **ChemoInteract Framework:** Designed and developed a modular PyTorch library unifying **11 published deep learning models** (GNNs, MLPs, Co-Attention) for drug interaction, polypharmacy side-effect, and synergy prediction.
- **Modular Pipeline:** Implemented Template Method pattern via abstract `unpack()` method, reducing experiment training code from 100+ lines of boilerplate to a 1-line `pipeline()` call.

- **Compatibility Layer:** Engineered GPU compatibility patches for TorchDrug `PackedGraph` objects, resolving tensor transfer issues across PyTorch versions.
- **Benchmarking & Testing:** Integrated 5 auto-cached datasets with `@lru_cache` and built a 4-file `pytest` suite verifying forward passes and data structures.

4.2 ATS Keywords

PyTorch, Deep Learning, Graph Neural Networks (GNN), GCN, Drug-Drug Interaction (DDI), Molecular Representation Learning, RDKit, SMILES, Morgan Fingerprints, Co-Attention, Binary Cross-Entropy, ROC-AUC, scikit-learn, Python, TorchDrug, Unit Testing, Modular Software Architecture.

5 Future Roadmap Quick Cheat Sheet

ChemoInteract Quick Reference Cheat Sheet

Library Entrypoint: `from chemointeract import pipeline`
Execution: `results = pipeline(dataset="drugcombdb", model="deepsynergy", epochs=100)`
Core Design Patterns: Template Method (`unpack()`), Resolver Pattern (`class-resolver`), Caching (`@lru_cache`), Monkey-patch Shim (`compat.py`).
Key Datasets: DrugCombDB, DrugComb, TwoSides, DrugbankDDI, OncoPolyPharmacology.
Top Models: DeepSynergy (FC baseline), DeepDDS (GCN), MHCADDI (Cross-drug Attention), GCNBMP (FFT Circular Correlation).